> js0nnn

Embedded systems, IoT and edge AI | B.Tech CSE (AI & ML), VIT, class of 2027

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$ experience

Software Development Intern, SGF Labs Sep 2026 – Present

- Developing firmware for a LoRa (SX1262)-connected medical monitoring device on STM32, ESP32 and nRF52.
- Integrating embedded software with hardware for reliable, low-power system operation.
- Architecting modular software and developing CI pipelines for automated builds and testing.

Software & Embedded Systems Intern, Kivenetics Nov 2025 – Apr 2026

- Developed PySide/Qt apps for motorized wheelchair control, vital-sign visualization and data logging.
- Designed system architecture with Jetson Orin Nano, AWR6843AOPEVM, sensors, displays and interfaces.

AI & IoT Training and R&D Intern, Edsols Innovations Jun 2025 – Jul 2025

- Hands-on AI/IoT robotics with RPi 4B and Jetson Nano for motor control, obstacle avoidance and edge AI.

$ projects

AI-Enhanced Blood Sample Analysis Oct 2025 – Present

RPi 5, ESP32, FreeRTOS, Qt, OpenCV

- Retrofitted a laboratory centrifuge with an RPi–ESP32 distributed control architecture.
- Implemented FreeRTOS motor control, PID speed regulation, encoder-based positioning and peripheral control.
- Achieved under 1° rotor precision, 20% improved speed regulation and soft-start control.

ADE9000 Power Quality Monitor Jan 2026 – May 2026

STM32H755, ADE9000, MQTT, Qt

- Developed STM32H755 firmware for power-quality monitoring with dual-core processing and SPI DMA.
- Implemented UART/DMA and MQTT telemetry for real-time monitoring and communication.
- Achieved 1% voltage and 5% current measurement error; hand-assembled the SMD ADE9000 circuit.

Context-Configurable AI Surveillance System Dec 2025 – Feb 2026

YOLO, ByteTrack, OpenCV, Qt

- Developed a context-based rule engine for activity-aware, real-time alerts.
- Processed 30 FPS feeds at 25–28 FPS single-camera and 15–20 FPS dual-camera rates.

Drowsiness Detection Seat for TATA Jun 2024 – Jun 2025

STM32F407, AWR1642BOOST, RPi 5

- Streamed flex-sensor, VL53L0X and AWR1642BOOST data via STM32F407 to RPi 5.
- Implemented a rule-based decision engine on RPi 5 for drowsiness detection and alerts within 1500–2000 ms.
- Developed T-SIM7600G-H based SOS SMS alerts with GPS location and haptic/sound feedback.

$ publications

- E-Sentinel: A context-aware emergency response platform with offline SOS support and neutrosophic fall detection, *Elsevier SoftwareX*, 2026.
- Grid Power 2.0: Artificial Intelligence (AI) and Internet of Things (IoT) Optimizations in Smart Power Grids, *IEEE AIIoT*, 2024.

$ skills

Languages	Java, C/C++, Python, SQL, QML
Frameworks/Libraries	Qt (PyQt/PySide), OpenCV, YOLO, FreeRTOS
Embedded Platforms	STM32, ESP32, RPi (3B/4B/5B), NVIDIA Jetson Nano, TI mmWave (AWR1642/AWR6843)
Embedded Interfaces	UART, SPI, I2C, GPIO, ADC, PWM, Timers, DMA, EXTI
Developer Tools	Git/GitHub, STM32CubeIDE, ESP-IDF, Linux (Debian-based), Fusion 360, VS Code